



PROJECT FOR DATA ANALYSIS – 1

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ACADEMIC STRESS AND MENTAL HEALTH:

SUMMARY

This research entitled "Academia's Toxic Stress – Breaking Students' Mental Health" gives a comprehensive analysis of the academic stress epidemic among university students. Composed of four parts, the research explores students' daily routines, levels of stress, institutional support, and mental health through descriptive statistics, frequency and percentage distributions, and cross-tabulations. The results show that students, especially 19–22 years old, are experiencing chronic stress because of poor sleep, excessive screen time, and insufficient rest or breaks. The majority of students experience high to extreme stress, emotional exhaustion, and inadequate time management. Institutional mental health care is rated as insufficient, and most students do not attend counseling because of stigma or lack of awareness. In the third part, variations by field of study, gender, and coping mechanisms were examined, wherein passive coping prevails, and support systems are utilized inconsistently. The statistical assessment of the fourth part indicates high scores for anxiety, low self-assurance in stress management, and negative correlations of burnout and institutional support. Finally, this research points to the critical necessity of universities to focus on mental health, implement structured coping programs, and cultivate a culture that places emotional well-being and academic excellence equally.

1st Part: Student Lifestyle and Daily Routines

Summary:

Students are suffering from:

- Chronicstress
- Sleep deprivation
- Digital fatigue
- Frequent feelings of being overwhelmed

Most:

- Do not take sufficient breaks
- Turn in late work due to stress
- Underuse counseling
- Report having 0–6 rest days per month

Key Insights:

- Age Group: Most are aged 19–22 years
- Sleep Patterns: Most students have 6–8 hours, some as little as 4 hours
- Study Time: Varies from 5 to 10+ hours per week
- Breaks: Some students take breaks; others study without any, increasing risk of burnout
- Class Attendance: Normally 2–3 classes per day
- Screen Usage: Academic screen time 2–6 hours per day
- Late Assignments: Often delayed due to stress
- Overwhelm Frequency: Students overwhelmed 2–3 days per week
- Counseling Use: Most have never gone to a session
- Rest Days: Many report only 0–6 rest days per month

Conclusion:

Universities need to:

- Increase mental health services
- Support taking breaks
- Cultivate a culture of balance and wellbeing

Questions & Scales:

1. How old are you (in years)? – Scale
2. On average, how many hours do you sleep at night? – Interval
3. How many hours do you study per week? – Interval
4. How many breaks do you take in a study session? – Interval
5. On average, how many classes do you go to per day (Monday - Friday)? – Interval
6. On average, how many hours per day do you spend using digital screens for schoolwork? – Interval

7. How many assignments were you late in turning in this semester because you were stressed out? – Interval

8. How many days a week do you feel overwhelmed because of school work? – Interval

9. How many mental health counseling sessions have you attended this semester? – Interval

10. How many rest days do you allow yourself each month? – Interval

FORMULAS IN PART 1

1. Mean (Average):

The middle value of a data set.

Formula:

Mean = Sum of all data points divided by the count of data points

Where:

- x_i = every data point
- n = count of data points

2. Median:

Middle value when data is sorted in order.

- if the count of values (n) is odd:

Median = Value at position of $(n + 1) / 2$

- if the count of values (n) is even:

Median = Average of the values at positions n divided by 2 and $(n$ divided by 2) + 1

3. Mode:

The value that appears most often in the dataset.

4. Range:

Difference between highest and lowest values.

Formula: $\text{Range} = \text{Highest value} - \text{Lowest value}$

5. Sample Variance (s^2):

Measure of the extent to which data points vary from the mean.

Formula:

Sample Variance = Total of squared difference between each and the mean divided by (num values - 1)

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6. Standard Deviation (s):

Sample variance square root.

Formula:

Sample Variance = Total of squared difference between each and the mean divided by (num values - 1)

7. Interquartile Range (IQR):

Spread of the middle 50% of values.

Formula:

$$\{\text{IQR}\} = Q3 - Q1$$

Where:

- $Q1 = 25\text{th percentile (lower quartile)}$
- $Q3 = 75\text{th percentile (upper quartile)}$

2nd Part: Emotional & Psychological Impact

Summary:

Students indicate high to extreme levels of academic stress, which result in both emotional and physical symptoms.

Patterns of note are:

- Emotional Exhaustion: Students frequently or always feel emotionally exhausted
- Physical Symptoms: Frequently report headaches and fatigue, from mild to severe
- Procrastination: Anxiety-driven, perpetuating the cycle of stress
- Post-Assignment Anxiety: Stress does not dissipate after submissions
- Poor Time Management: Consistently poor or very poor
- Lack of Self-Care: Rarely place priority on mental and physical health
- Peer Support: Variable; frequent to none
- University Perception: Mental health as not important or only somewhat important

Key Insights:

- Anxiety and burnout are long-standing and strongly embedded in academics
- Institutional support is not seen as sufficient
- Students are conscious of stress but cannot manage or reduce it

Conclusion:

There is an immediate need to:

- Enhance awareness and accessibility of mental health resources
- Promote self-care and stress-management practices
- Develop a culture of peer support and faculty support

Questions & Scales:

11. Rate your overall academic level of stress. – Ordinal
12. How often do you experience feelings of emotional exhaustion from academic pressures? – Ordinal
13. How strong are your physical symptoms of stress (e.g., headache, fatigue)? – Ordinal
14. How often do you procrastinate because you worry? – Ordinal
15. How worried do you feel once you've finished major projects? – Ordinal
16. How would you rate your ability to manage academic pressure? – Ordinal
17. How would you rate how well you manage time throughout the semester? – Ordinal
18. How often do you engage in self-care throughout the semester? – Ordinal
19. How much support do you receive from your peers during stressful times of academics? – Ordinal
20. To what extent do you believe your university places a value on menstrual health? – Ordinal

FORMULAS IN PART 2:

1. Frequency Distribution

A table that indicates how frequently each unique value in a data set appears.

- Frequency (f): Number of occurrences for a specified value or class interval.

There is no formula for frequency itself—it is just counted from the data.

2. Percentage Distribution

This translates frequencies into percentages to make comparisons between different categories easier.

% distribution = $F / \text{total frequency} * 100$

Where:

- Frequency = number of observations within a class/category
- Total Frequency = aggregate number of all the observations

3rd Part: Demographics, Stressors & Coping

Summary:

This part looks at the background of students, main stressors, and how they cope with stress in their studies.

Key Observations:

- Year of Study:

Most high levels of stress are reported by 1st and 2nd year students

- Majors:

Business, Sciences, and Engineering are the most common fields, typically with high academic requirements

- Gender:

Male and female students both report stress, with slight differences in coping

- Institution Type:

Same patterns of stress seen across public and private universities

- Primary Stressors Include:
 - Exams and assignments
 - Personal expectations
 - Lack of time and time pressure
 - Coping Strategies:
 - Passive methods are prevalent (e.g., sleep, withdrawal)

- Active methods involve talking to friends and hobbies
- Some students have no coping strategies
 - Stress Relief Activities:
- Music, gaming, meditation, and hanging out with friends
 - Study Preferences:
- Mainly individual study
- Then online resources
- Then group study
 - Mental Health Support Usage:
- Several students do not use any official support
- Some use support from peers or online counselling
 - Social Media Usage:
- Most popular sites: Instagram and YouTube
- Some students steer clear of social media because of its adverse effect on stress

Conclusion:

Students experience varied but severe stressors across fields and backgrounds.

There is a clear need for improved access to coping resources, especially formal mental health services.

Questions & Scales:

21. What is your current year of study? – Ordinal
22. What is your academic major or field of study? – Nominal
23. What is your gender identity? – Nominal
24. What type of institution are you enrolled in (e.g., private, public)? – Nominal
25. What are your top sources of academic stress? – Nominal
26. What are your coping mechanisms for stress? – Nominal
27. What do you do to relax and relieve stress? – Nominal
28. What is your favorite way of studying? – Nominal
29. What mental health services do you utilize, if any? – Nominal
30. How does social media influence your academic stress? – Ordinal

FOMULAS FOR 3 part:

1. Frequency Distribution

This is a table that shows how often each different value in a set of data occurs.

- Frequency (f): The number of times a particular value or class interval occurs.

There's no formula for frequency itself—it's simply counted from the data.

2. Percentage Distribution

- This converts frequencies into percentages to compare different categories more easily.

% distribution = $F \text{ divided by total frequency multiply by } 100$

Where:

- Frequency = number of observations in a class/category
- Total Frequency = total number of all observations

4th Part: Quantitative Analysis and Statistical Correlations

Summary:

This section provides statistical analyses of student feedback, which emphasizes major trends in stress levels, emotional health, and the influence of support systems.

Key Metrics & Insights:

- Finals Week Stress:
 - Mean $\approx$ 3.5, Median = 4
 - Reflects high pressure during exam times
- Anxiety & Burnout:
 - Mean around 3.0 to 3.2
 - Most students score 4 or 5, reflecting chronic stress
- Academic Pressure:
 - Mean $\approx$ 3.4
 - 25% of students scored it 5 (extreme)
- Stress Management Confidence:
 - Mean $\approx$ 2.8
 - 20% rate themselves (very low confidence)
- Overall Emotional Well-Being: Mean $\approx$ 2.9
 - 30% of students rated it 1 or 2, showing poor mental health
- Sleep Quality During Exams:
 - Median = 2
 - 40% rated their sleep quality as 1 (very poor)
- Satisfaction with Mental Health Support: Mean $\approx$ 2.3
 - 35% rated their satisfaction 1
- Energy & Focus: Energy Mean $\approx$ 2.7
 - Focus Mean $\approx$ 3.0

Correlations:

- Stress and Anxiety: $r \approx 0.6$ – Strong positive correlation
- Sleep and Concentration: $r \approx 0.5$ – Moderate positive relationship
- Support Satisfaction and Burnout: $r \approx -0.4$ – Reverse relationship

Skewness:

- Negative skew towards higher levels of stress and anxiety, which is to say that extreme cases are more prevalent than mild ones

Implications:

- Academic stress is systemic, not the product of a few individuals
- More than 50% of students employ no formal mental health support
- Passive coping (e.g., sleep, withdrawal) is rampant
- Institutional interventions are needed

References:

- Extend and de-stigmatize counseling services
- Apply stress-reduction initiatives (e.g., workshops, breaks, peer support)
- Normalize discussing mental health
- Encourage active coping strategies

Questions & Scales:

31. How stressed do you feel during finals week? – Ordinal
32. How often do you experience academic anxiety? – Ordinal
33. How often do you feel burnout due to academic demands? – Ordinal
34. How confident are you in managing your academic stress? – Ordinal
35. How would you rate your overall emotional well-being? – Ordinal
36. How would you rate your sleep quality during exams? – Ordinal
37. How satisfied are you with your institution's mental health support? – Ordinal
38. How would you rate your energy levels during the semester? – Ordinal
39. How well are you able to focus on academic tasks? – Ordinal

40. How often do you use formal mental health services (e.g., counseling, therapy)? – Ordinal

FORMULAS FOR 4 part:

1. Mean (Average):

The average of a data set.

Formula:

Mean = Total sum of all values divided by the number of values

Where:

- x_{t_i} = individual data point
- n = number of data points

2. Median:

The middle value when data is listed in order.

- If the number of values (n) is odd:

Median = Value at position of $(n + 1) / 2$

- If the number of values (n) is even:

Median = Average of values at the positions n divided by 2 and $(n$ divided by 2) + 1

3. Mode:

The value most likely to occur in the set.

4. Range:

The range is the difference between the biggest and smallest values.

Range = Highest value - Lowest value

5. Sample Variance (s^2):

A measure of the spread of the data points from the mean.

Formula: Sample Variance = Sum of the squared differences between each value and the mean, divided by (number of values minus 1)

.6. Standard Deviation (s):

The square root of the sample variance.

Formula:

Sample Variance = Sum of the squared differences between each value and the mean, divided by (number of values minus 1)

7. Interquartile Range (IQR):

The range of the middle 50% of data.

Formula:

$$\{IQR\} = Q3 - Q1$$

Where:

- Q1= 25th percentile (lower quartile)
- Q3= 75th percentile (upper quartile)

8. Formula for Percentile:

Percentile = The value below which a given percentage of the data falls

To calculate the k-th percentile:

Position = (k divided by 100) multiplied by (number of values + 1)

How we applied these formulas in excel:

1.Mean:

= AVERAGE(select range)

This formula add all values that are in range and divide them by number of cells

2.Median:

= MEDIAN(select range)

- Median is the middle value If there is an odd number of values, it returns the middle one.

- If there is an even number of values, it returns the average of the two middle values.

3.Mode;

=MODE.SNGL(RANGE)

Mode is the frequently occurring value in data set.

4.Range:

=MAX(RANGE) - MIN(RANGE)

To find range we will subtract maximum from minimum value in data set

5.Sample variance:

=VAR.S(A2:A51)

A larger variance means the data points are more spread out from the mean; a smaller variance means they are more clustered around the mean.

6.Standard deviation:

=STDEV.S(A2:A51)

Standard deviation is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean, while a high standard deviation indicates that the values are spread out over a wider range.

7.INTERQUARTILE RANGE:

=PERCENTILE.EXC(RANGE, 0.25)

=PERCENTILE.EXC(RANGE, 0.75)

=PERCENTILE.EXC(RANGE, 0.75) - PERCENTILE.EXC(RANGE, 0.25)

The Interquartile Range (IQR) is a measure of statistical dispersion, or in simple terms, it shows how spread out the middle 50% of the data is. It is the range between the first quartile (Q1) and the third quartile (Q3).

8.PERCENTILES;

Percentiles are values that divide a dataset into 100 equal parts, with each part representing 1% of the data. In other words, a given percentile indicates the position below which a certain percentage of observations fall.

=PERCENTILE.EXC(RANGE, 0.25)

=PERCENTILE.EXC(RANGE, 0.50)

=PERCENTILE.EXC(RANGE, 0.75)

9.OUTLIERS DETECTION:

Outlier detection involves identifying data points that differ significantly from the majority of a dataset. Outliers can affect statistical analyses and may indicate variability, measurement errors, or novel phenomena.

=(A2 - AVERAGE(range)) / STDEV.P(range)

10.skewness;

Skewness is a statistical measure that describes the asymmetry of a distribution around its mean.

=SKEW(RANGE)

11. CORRELATION ANALYSIS:

Correlation analysis is a statistical technique used to measure and describe the strength and direction of a relationship between two variables.

=CORREL(Range of column A ,Range of column B)

12.poison probability distribution:

Poisson probability refers to the probability of a given number of events occurring in a fixed interval of time or space, assuming these events happen independently and at a constant average rate.

=POISSON.DIST(2, AVERAGE(RANGE), TRUE)

13.COEFFICIENT OF VARIANCE:

The coefficient of variation (CV) is a standardized measure of relative variability. It shows the extent of variability in relation to the mean of the data, and it is especially useful for comparing the variation between datasets with different units or means.

$$= \text{STDEV.S}(\text{RANGE}) / \text{AVERAGE}(\text{RANGE})$$

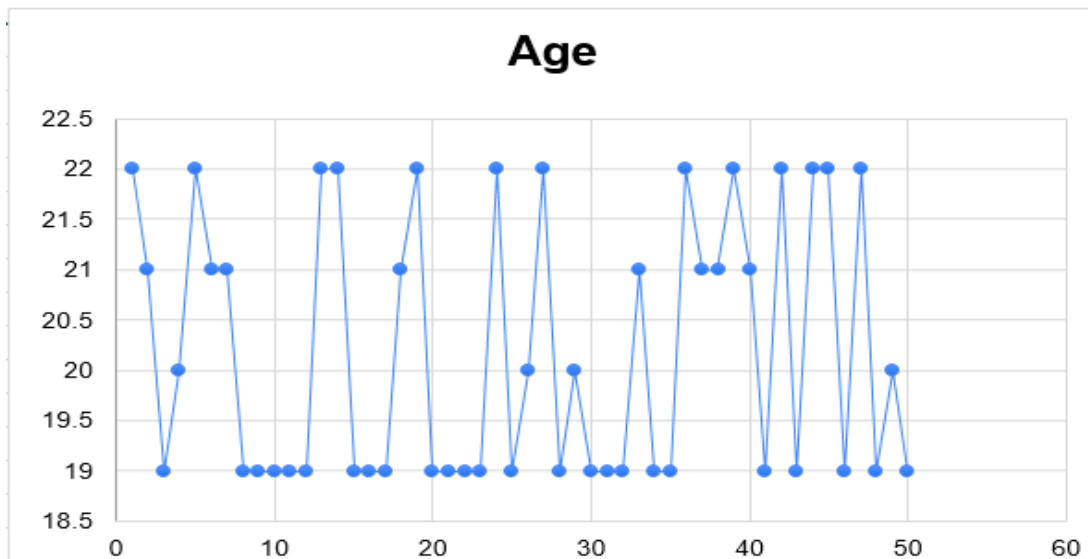
GRAPHS AND PLOTS:

1. SCATTER PLOT:

1. Enter your data in two columns (e.g., X values in column A, Y values in column B).
2. Select the data (both columns).
3. Go to the Insert tab on the Ribbon.
4. In the Charts group, click on the Insert Scatter (X, Y) or Bubble Chart icon.
5. Choose a Scatter type (e.g., just points or points with lines).
6. Excel will insert the scatter plot into your worksheet.

This figure shows scatter plot for

QUESTION: HOW ARE YOU IN YEARS?



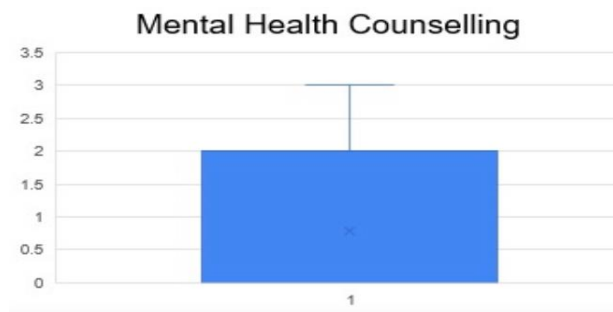
Y AXIS SHOWS NUMBER OF RESPONSES AND X AXIS SHOWS

X AXIS SHOWS INDEPENDENT VARIABLE

2:BOX PLOT:

1. Minimum – the smallest data point (excluding outliers)
2. First Quartile (Q1) – the median of the lower half of the dataset
3. Median (Q2) – the middle value
4. Third Quartile (Q3) – the median of the upper half
5. Maximum – the largest data point (excluding outliers)

This box plot shows minimum point is 0 and maximum point is 3



3: Histograms:

- X-axis: Represents intervals (bins) of data.
- Y-axis: Represents the frequency (count) of data points in each bin.
- Bars: Each bar shows how many data points fall into that bin.
- No gaps between bars (unless there's a gap in data), since the data is continuous.

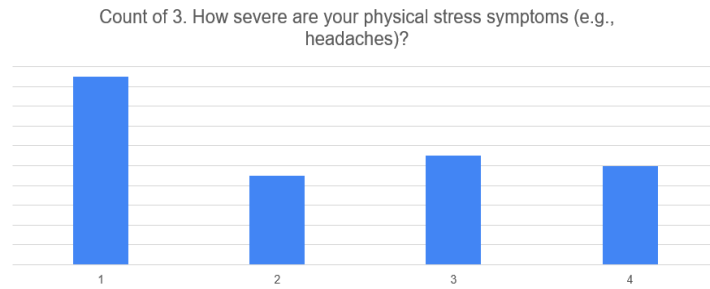


Y axis shows frequency and x axis population size

4.BAR CHART:

- X-axis (horizontal): Represents the values (numerical data).
- Y-axis (vertical): Represents the categories (labels or items being measured).

This given barchart shows students who have physical stress symptoms



CROSS TABULATION:

Cross tabulation (or cross-tab) is a method used in statistics to analyze the relationship between two or more categorical variables by displaying the data in a table format. Each cell in the table shows the frequency or count of observations that fall into the corresponding categories of the variables.

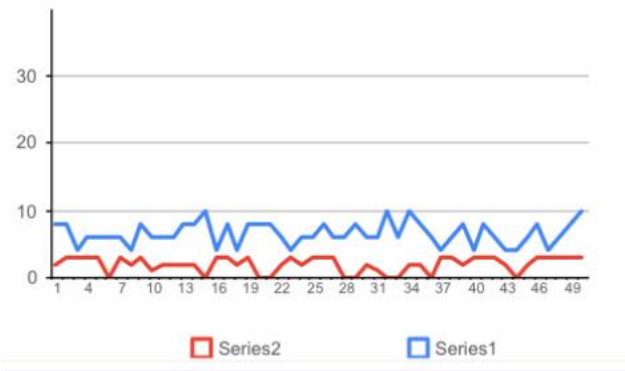
This given Cross tabulation table shows Major of population and the year in which they are studying.

Count of 3. What is your gender?	Year					
Major	1st year	2nd year	3rd year	4th year	Grand Total	
Business	6	7	3	2	18	
Engineering	2	6		5	13	
Sciences	5	9	2	3	19	
Grand Total	13	22	5	10	50	

PROBLEM STATEMENT:

Is there any relation between number of hours students sleep and the number of days they feel overwhelmed due to academic workload?

This **scatter plot** shows relationship between column A and column B



This **Pie chart** shows relation between column A and column B

